

# **Technical Cybersecurity Report**

**Analysis using splunk to record event logs  
(step-by-step guide on downloading and installing  
Splunk, including clearing event viewer security  
logs and creating a dashboard on splunk)**

**By**

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## REPORT OUTLINE

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## **1. Executive Summary**

### **Overview:**

This security analysis harness splunk as a Security Information and Event Management (SIEM) system to observe and track Windows security logs. Splunk Enterprise was downloaded, installed, and configured on a Windows system to collect event logs, clear historical security logs and utilized to generate a real-time security dashboard. This initiative is to improve security event visibility and incident detection capabilities and response.

### **Key Findings:**

Splunk Enterprise was successfully installed and configured to process Windows security logs. Historical event viewer security logs were cleared to form a baseline for fresh monitoring, a custom dashboard was created to provide visibility for key metrics such as login attempt and system changes. No immediate threats were identified, but the setup demonstrated Splunk's ability to detect anomalies and provide actionable insights into system activity.

## **2. Background and Objectives**

### **Project Context:**

The project aimed to enhance the organization's cybersecurity posture by deploying a SIEM solution to track system logs, detect potential security threats, and integrate Windows Event Viewer security logs into a centralized monitoring platform.

### **Objective of the Tool Use:**

The main objective is to use splunk to ensure proactive identification of suspicious activities, clear outdated Event Viewer logs and establish a baseline for security monitoring. Splunk was mobilized to install and configure a SIEM system and custom a dashboard for visualising security activities .

## **3. Methodology**

### **3.1 Tool Configuration**

#### **Splunk Configuration:**

##### **Download Splunk:**

- Go to the official Splunk website (splunk.com).
- Sign up or log in to download Splunk Enterprise (free trial or licensed version).
- Choose the correct package for your OS (e.g., .tgz for Linux, .msi for Windows).
- Download the .msi file from the Splunk website (e.g., splunk-9.2.0-x64-release.msi).

## Install Splunk:

- Double-click the .msi file to launch the installer.
- Ran the .msi installer as an administrator.
- Click "Next" through the wizard:
  - Accept the license agreement.
  - Choose the installation directory (default is C:\Program Files\Splunk).
  - Select "Local System" for the service account (or a specific account if required).
- Click "Install" and wait for the process to finish.
- Check the box to "Launch browser with Splunk" when prompted, then click "Finish."

## Set Admin Credentials:

- The browser opens to <http://localhost:8000>.
- Set an admin username and password when prompted.

## 3.2 Execution Process

### Clearing Event Viewer Security Logs

- Open the splunk app or the web interface,
- Go to setting click on data input and select local event log collection,
- Select application system and security logs from the available logs and move to the selected log and save changes
- Click on search and reporting on the splunk homepage
- Click on the windows button on the system, type in eventviewer and hit the enter button
- Select application, system and security then clear log to clear the windows event leaving only one log which is the recent log (log 1102)
- Return to the splunk and type \* on the search. This will display all the logs that has gone through the system, security and application.as were initially configured in the setting
- Move the cursor to the host machine, Left click and add to search. it filter the event to only logs from the host machine.
- proceed to the WinEventLogSecurity,left click and add to search
- Look for event code, left click and add to search
- Go the website <http://Ultimatewindowsecurity.com>.
  - Click on security log
  - Navigate to Encyclopedia for code interpretation,
  - Edit the code in the search query and input 1102 and click enter

### **Dashboard Creation:**

Copy the search query and save on the system notes for later use .

- Go to the right side of the search button on the splunk homepage to edit the time frame to any desired time.
- click on create a tableview , unclick the raw data to present a clean less detailed information
- Go to dashboard, click on create new dashboard, fill in your desired details , select dashboard studio, choose grid and click to create to save the selected options
- Select search 1 from filter, paste and run the saved search query on the spl query apply and close
- On the spl query input | fields - \_bkt \_cd \_indextime to filter out the bk,cd and indextime from the display, apply, close and click on save to save the data
- Click on dashboard navigate to the saved file and click on it to open it, click on action navigate to set as homepage dashboard for easy access
- logs when cleared the information will automatically appear on the dashboard.

### **3.3 Monitoring and Analysis**

#### **System Monitoring:**

Splunk logged all changes post-clearing thereby creating a fresh baseline , it also captured real-time events from Event Viewer including log clearing action (EventCode=1102).

#### **Network Monitoring:**

Though this analysis is focused on system logs splunk's network monitoring was enabled to invigilate outbound connections, ensuring a wider visibility to threat vectors.

## **4. Findings and Analysis**

### **4.1 Indicators of Compromise (IOCs)**

#### **Detected IOCs:**

Although no malicious IOCs were identified during the analysis as it is based on splunk installation and clearing event log to establish a fresh baseline for future security monitoring

### **4.2 Behavioral Analysis**

#### **System Actions:**

Splunk logged normal system behaviour post-installation such as successful event log clearing actions (EventCode 1102). No suspicious activity was observed during the installation process and analysis.

**Persistence Mechanisms:**

No persistence mechanisms were analyzed, as this report focuses on tool installation and deployment rather than malware investigation.

**4.3 Risk and Impact Assessment****Potential Impact:**

Without an active SIEM monitoring system like Splunk, uncleared logs could mask new threats, and lack of a dashboard could prevent detection of incidents like unauthorized access. If unmonitored, such events could lead to system compromise or data breach.

**5. Recommendations****5.1 Immediate Remediation Actions**

- Audit your system regularly to monitor user and administrator access, as well as other activities that could tip you off to unsafe practices or security breaches.
- Set up users and configure roles and capabilities to control user access
- Single sign-on with multi-factor authentication (MFA) using Duo MFA.

**5.2 Long-Term Mitigation****Security Measures:**

- Regularly patch Splunk to address vulnerabilities.
- Configure real-time automated alerts for anomalies to detect potential threat attempts and privilege escalation
- Expand the dashboard to include network traffic metrics and integrate threat intelligence feeds to flag known malicious IPs or domains.

**6. Conclusion****Summary of Findings:**

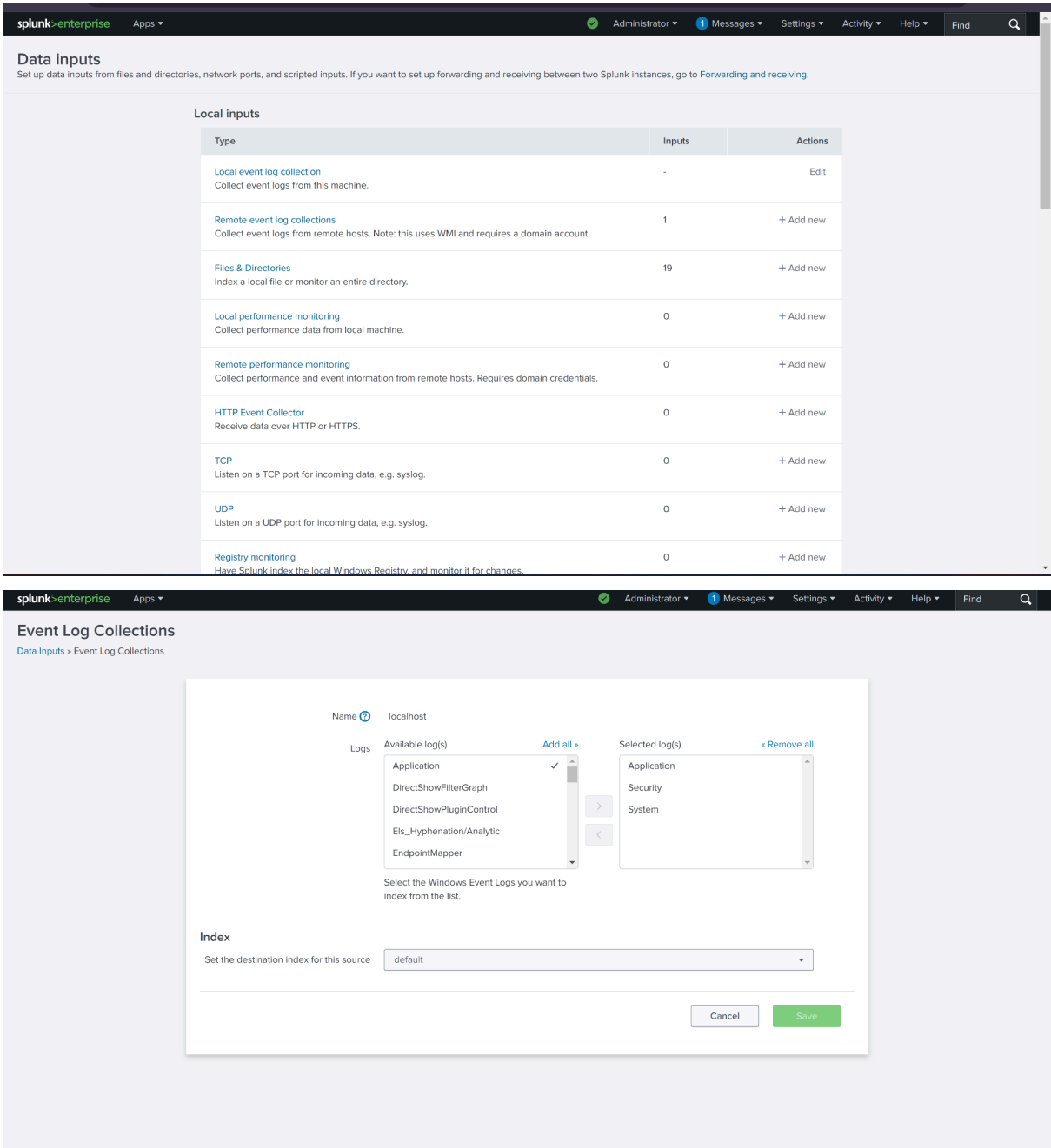
Splunk was successfully downloaded, installed and configured to monitor Windows's security event logs. Historical event logs were clear to foster a fresh baseline for future logs and security monitoring, a security dashboard was created to ensure capturing of onward security and Windows log events.

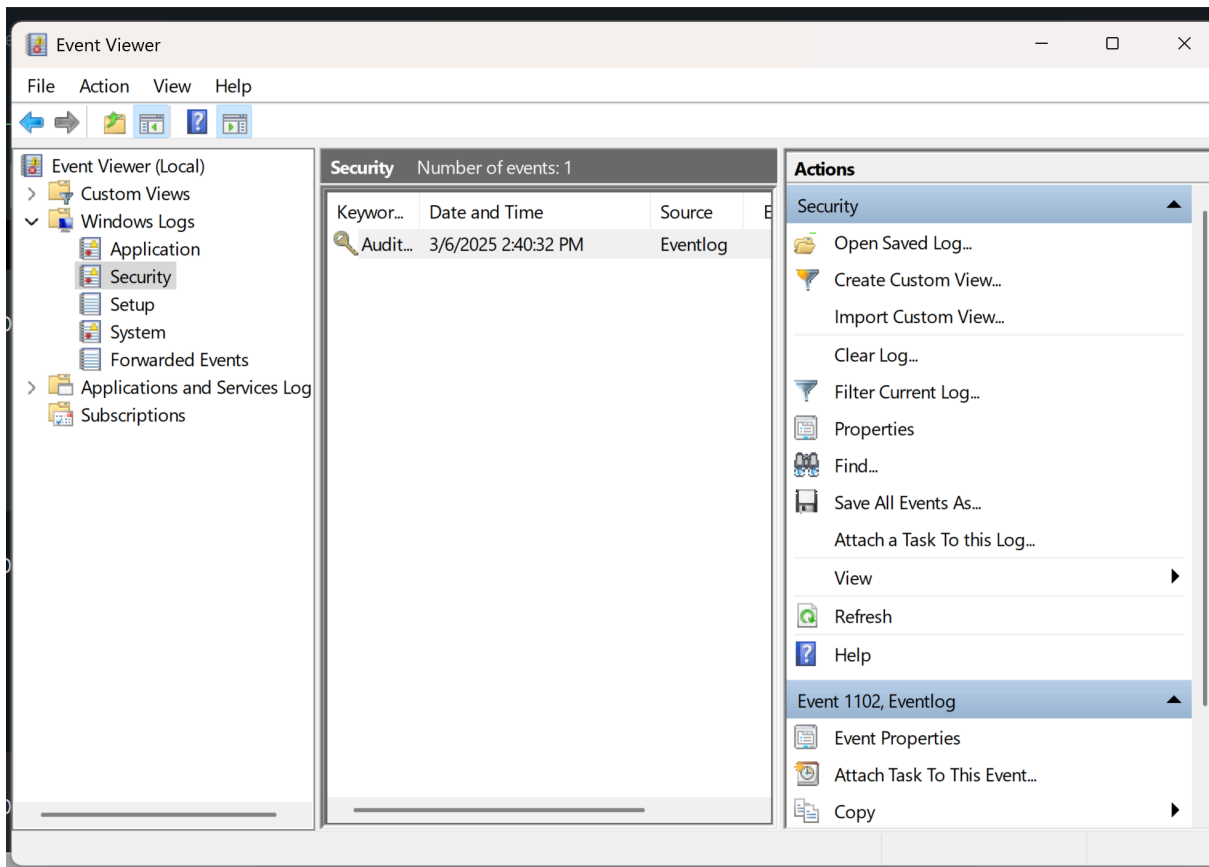
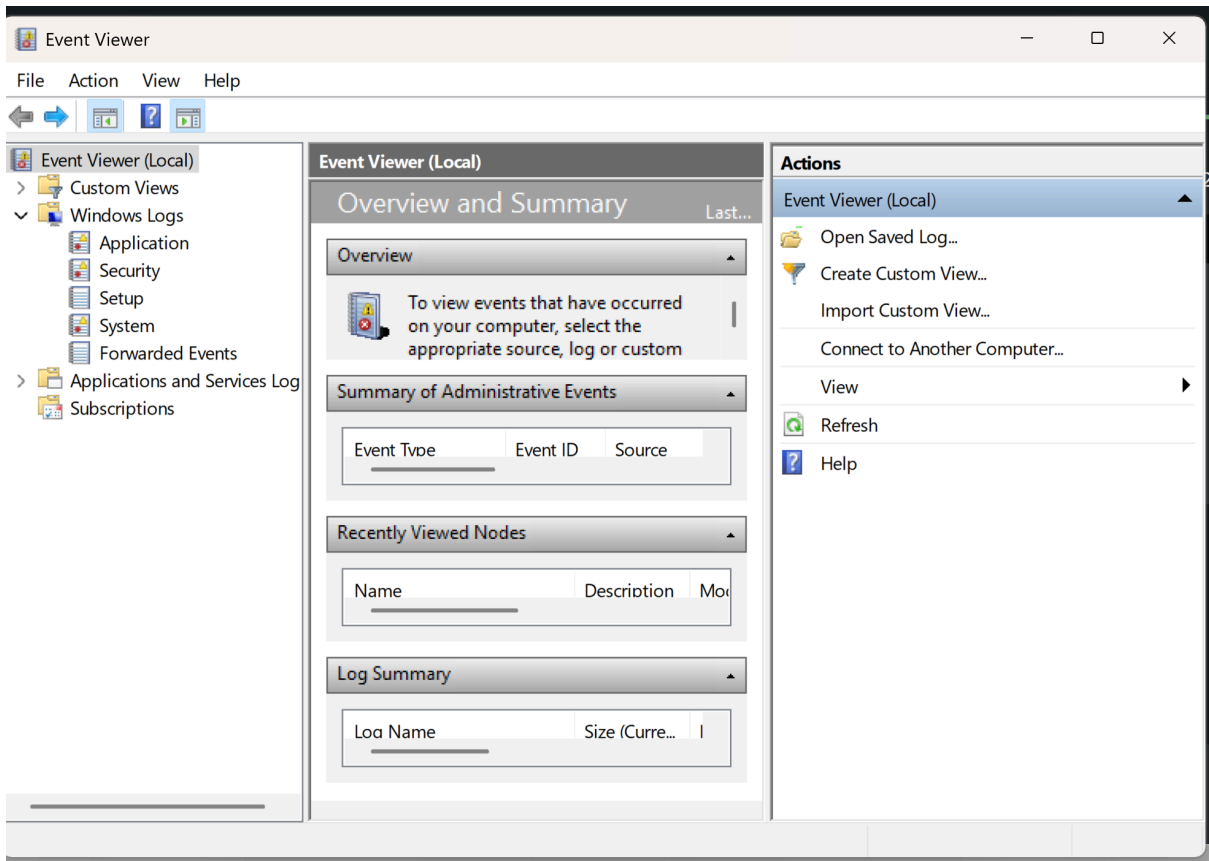
No threat was discovered in the course of the analysis but is recommended to set up a user access control system to limit and eliminate unauthorized access to the Splunk Enterprise as unauthorized privilege could alter configuration. Also regular

patch up of the splunk is encouraged and configuring and set up of automated alert for anomalies should be adhered to for an advance security measure and proactive mediation.

## Appendix: Additional Data

### Raw Logs and Tool Outputs:







splunk-enterprise Apps ▾ Administrator ▾ Messages ▾ Settings ▾ Activity ▾ Help ▾ Find

Search Analytics Datasets Reports Alerts Dashboards Search & Reporting

### New Search

Save As Create Table View Close

host="DESKTOP-NAGLOS4" Last 24 hours

10,273 events (3/5/25 2:00:00.000 PM to 3/6/25 2:43:28.000 PM) No Event Sampling

Events (10,273) Patterns Statistics Visualization

Timeline format Zoom Out Zoom to Selection Deselect 1 hour per column

Format Show: 20 Per Page View: List

Hide Fields All Fields

SELECTED FIELDS

- host 1
- source 3
- sourcetype 3

INTERESTING FIELDS

- Account\_Domain 4
- Account\_Name 8
- ComputerName 1
- date\_hour 23
- date\_mday 2
- date\_minute 60
- date\_month 1
- date\_second 60
- date\_wday 2
- date\_year 1
- date\_zone 1
- EventCode 67
- EventType 4

| i | Time                     | Event                                                                                                                                                                                                                                       |
|---|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| > | 3/6/25<br>2:43:01.000 PM | 03/06/2025 02:43:01 PM<br>LogName=Application<br>EventCode=16394<br>EventType=4<br>ComputerName=DESKTOP-NAGLOS4<br>Show all 12 lines<br>host = <b>DESKTOP-NAGLOS4</b> source = WinEventLog:Application sourcetype = WinEventLog:Application |
| > | 3/6/25<br>2:40:32.000 PM | 03/06/2025 02:40:32 PM<br>LogName=Security<br>EventCode=1102<br>EventType=4<br>ComputerName=DESKTOP-NAGLOS4<br>Show all 17 lines<br>host = <b>DESKTOP-NAGLOS4</b> source = WinEventLog:Security sourcetype = WinEventLog:Security           |
| > | 3/6/25<br>2:35:46.000 PM | 03/06/2025 02:35:46 PM<br>LogName=Security<br>EventCode=4672                                                                                                                                                                                |



February 2025 Patch Tuesday

\*Patch Tuesday - Four Zero Days: Average

User name:

Password:

[Login](#) / [Forgot?](#)

[Register](#)

[Security Log](#)
[Windows](#)
[SharePoint](#)
[SQL Server](#)
[Exchange](#)
[Training](#)
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## Resources for the Cryptic Windows Security Log



Free Quick Reference Chart...



Free Windows Event Collection Management



Free Webinars...



Online Encyclopedia...



Free Training



The Book

February 2025  
Patch Tuesday

\*Patch Tuesday - Four Zero Days:

User name:

Password:

Login

Forgot?

Register

Ultimate IT  
SECURITY

Security Log

Windows

SharePoint

SQL Server

Exchange

Training

Tools

Newsletter

Webinars

Blog

Webinars

Training

Encyclopedia

Quick Reference

Book

Encyclopedia

• Event IDs

• All Event IDs

• Audit Policy

Go To Event ID:

Go

Security Log Quick Reference Chart

Download now!

Post

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Share

Windows Security Log Events

All Sources

Windows Audit

SharePoint Audit (LOGbinder for SharePoint)

SQL Server Audit (LOGbinder for SQL Server)

Exchange Audit (LOGbinder for Exchange)

Sysmon (MS Sysinternals Sysmon)

Windows Audit Categories:

All categories

Subcategories:

All subcategories

Windows Versions:

All events

Win2000, XP and Win2003 only

Win2008, Win2012R2, Win2016 and Win10+, Win2019

Category: All

Windows 1100 The event logging service has shut down

Windows 1101 Audit events have been dropped by the transport.

Windows 1102 The audit log was cleared

Windows 1104 The security Log is now full

Windows 1105 Event log automatic backup

Windows 1108 The event logging service encountered an error

Windows 4608 Windows is starting up

Windows 4609 Windows is shutting down

Windows 4610 An authentication package has been loaded by the Local Security Authority

Windows 4611 A trusted logon process has been registered with the Local Security Authority

Windows 4612 Internal resources allocated for the queuing of audit messages have been exhausted, leading to the loss of some audits.

Windows 4614 A notification package has been loaded by the Security Account Manager.

Windows 4615 Invalid use of LPC port

Windows 4616 The system time was changed.

Windows 4618 A monitored security event pattern has occurred

Windows 4633 A login failed because the user was not found in the local security database

splunk>enterprise

Apps

Administrator

Messages

Settings

Activity

Help

Find

Search

Analytics

Datasets

Reports

Alerts

Dashboards

Search & Reporting

New Search

Save As

Create Table View

Close

\* host="DESKTOP-NAGL054" source="WinEventLog:Security" "EventCode=1102"

Last 24 hours

Q

1 event (3/5/25 2:00:00.000 PM to 3/6/25 2:51:51.000 PM)

No Event Sampling

Job

||

→

↑

↓

Smart Mode

Events (1)

Patterns

Statistics

Visualization

Timeline format

Zoom Out

+ Zoom to Selection

X Deselect

1 hour per column

Format

Show: 20 Per Page

View: List

< Hide Fields

All Fields

SELECTED FIELDS

a host 1

a source 1

a sourcetype 1

INTERESTING FIELDS

a Account\_Name 1

a ComputerName 1

# date\_hour 1

# date\_mday 1

# date\_minute 1

# date\_month 1

# date\_second 1

# date\_wday 1

# date\_year 1

# date\_zone 1

a Domain\_Name 1

# EventCode 1

# EventType 1

i Time Event

> 3/6/25 03/06/2025 02:40:32 PM

LogName=Security

EventCode=1102

EventType=4

ComputerName=DESKTOP-NAGL054

Show all 17 lines

host = DESKTOP-NAGL054 source = WinEventLog:Security sourcetype = WinEventLog:Security



splunk-enterpriseApps

AdministratorMessagesSettingsActivityHelpFind

SearchAnalyticsDatasetsReportsAlertsDashboards

Search & Reporting

DarkViewSave

task 12 B WORK LOGS

Enter dashboard description.

Global Time Range

Last 24 hours

New tab

New data source

Data source name

Search\_1

☐ Access search results or metadata

SPL query

Open in search

\* host="DESKTOP-NAGLOS4" source="WinEventLog:Security" "EventCode=1102"

Time range

InputStaticDefault

Time range set by dashboard source default value

\$global\_time.earliest\$ - \$global\_time.latest\$

Configure in source editor

Usage

1 visualization

0 inputs

0 chain searches

1 visualization will be updated

CancelApply and close

splunk-enterpriseApps

AdministratorMessagesSettingsActivityHelpFind

SearchAnalyticsDatasetsReportsAlertsDashboards

Search & Reporting

DarkViewSave

task 12 B WORK LOGS

Enter dashboard description.

Global Time Range

Last 24 hours

New tab

| _bkt                                        | _cd       | _inde...   | _raw                                                                                                                               | _serial | _sl             | _sour...             | _time                     | host            | index | lineco... | source               |
|---------------------------------------------|-----------|------------|------------------------------------------------------------------------------------------------------------------------------------|---------|-----------------|----------------------|---------------------------|-----------------|-------|-----------|----------------------|
| main~O~7E4999FE-3AE4-427E-9869-E215F259AD01 | 0:1037288 | 1741268432 | 03/06/2022 02:40:32 PM<br>LogName=Security<br>EventCod e=1102<br>EventType=4<br>Computer Name=DESKTOP-NAGLOS4<br>SourceName=Micros | 0       | DESKTOP-NAGLOS4 | WinEventLog:Security | 2025-03-06T14:00:00+01:00 | DESKTOP-NAGLOS4 | main  | 17        | WinEventLog:Security |

Configuration

General

Visualization type

Table

Title

Description

Data sources

Search\_1

Visibility

Data display

Color and style

Interactions

Source code

splunk>enterpriseApps

Apps

Find more apps

Search apps by name...

Search & Reporting

ATAudit Trail

Splunk Secure Gateway

Upgrade Readiness App

Hello, Administrator

Dashboard

task 12c log

Global Time RangeLast 24 hours

| _raw                                                                                                                                                 | _serial | _si                  | _source...           | _time                         | host            | index | linecount | source               | sourcet...           | splunk_...      |
|------------------------------------------------------------------------------------------------------------------------------------------------------|---------|----------------------|----------------------|-------------------------------|-----------------|-------|-----------|----------------------|----------------------|-----------------|
| 03/06/2025 02:40:32 PM<br>LogName=Security<br>EventCode=1102<br>EventType=4<br>ComputerName=DESKTOP-NAGLOS4<br>SourceName=Microsoft-Windows-Eventlog | 0       | DESKTOP-NAGLOS4 main | WinEventLog:Security | 2025-03-06T14:40:32.000+01:00 | DESKTOP-NAGLOS4 | main  | 17        | WinEventLog:Security | WinEventLog:Security | DESKTOP-NAGLOS4 |

splunk>enterpriseApps

Apps

Find more apps

Search apps by name...

Search & Reporting

ATAudit Trail

Splunk Secure Gateway

Upgrade Readiness App

Hello, Administrator

Dashboard

task 12c log

Global Time RangeLast 24 hours

| _raw                                                                                                                                                 | _serial | _si                  | _source...           | _time                         | host            | index | linecount | source               | sourcet...           | splunk_...      |
|------------------------------------------------------------------------------------------------------------------------------------------------------|---------|----------------------|----------------------|-------------------------------|-----------------|-------|-----------|----------------------|----------------------|-----------------|
| 03/06/2025 03:44:38 PM<br>LogName=Security<br>EventCode=1102<br>EventType=4<br>ComputerName=DESKTOP-NAGLOS4<br>SourceName=Microsoft-Windows-Eventlog | 0       | DESKTOP-NAGLOS4 main | WinEventLog:Security | 2025-03-06T15:44:38.000+01:00 | DESKTOP-NAGLOS4 | main  | 17        | WinEventLog:Security | WinEventLog:Security | DESKTOP-NAGLOS4 |

